

Physiology of Human Respiratory System

Outline

Respiratory System

- Functions
- Conducting zone
- Respiratory zone
- Lungs
- Gas transport
- Respiratory regulation

Physiology of Respiration

General objectives of Teaching

At the end of teaching students are expected to understand that:

1. The continuity of life requires a respiratory process that supplies O_2 and eliminates CO_2 continuously.
2. Respiration physiology is the basis of understanding diseases of the respiratory passages and the lungs.

Respiratory physiology

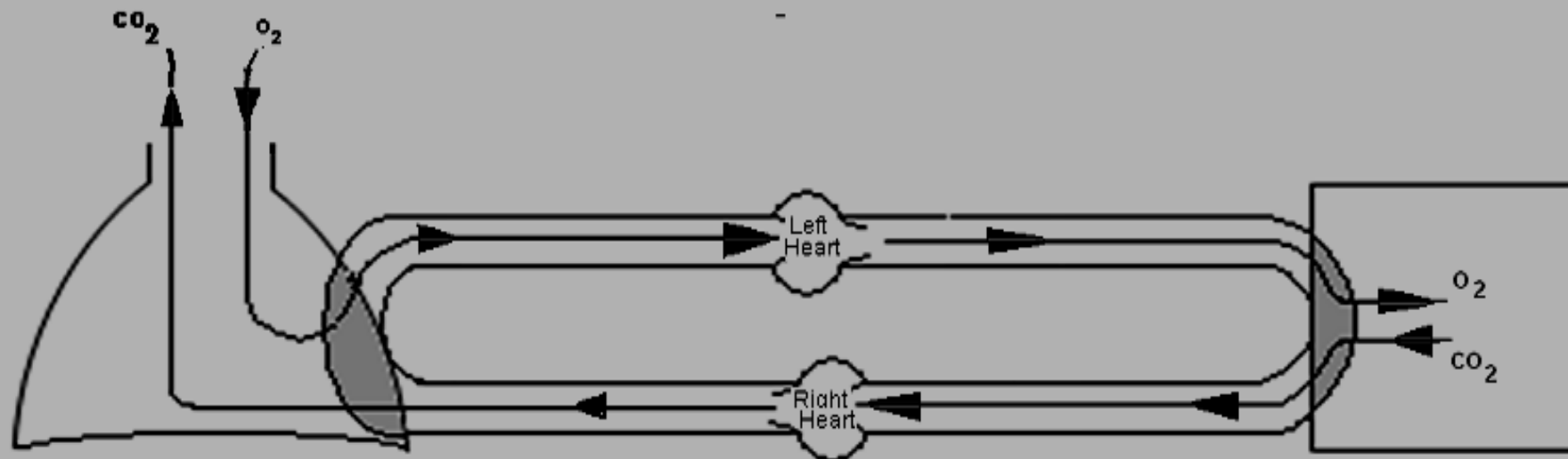
Basic functions of the respiratory system

- Gas exchange –supply oxygen to aerobic tissues in the body and remove carbon dioxide as a waste product.
- Regulation of acid-base balance:
- Filtration of blood clot
- Conversion of ang-I to ang-II by the action of ACE
- Involved in defence

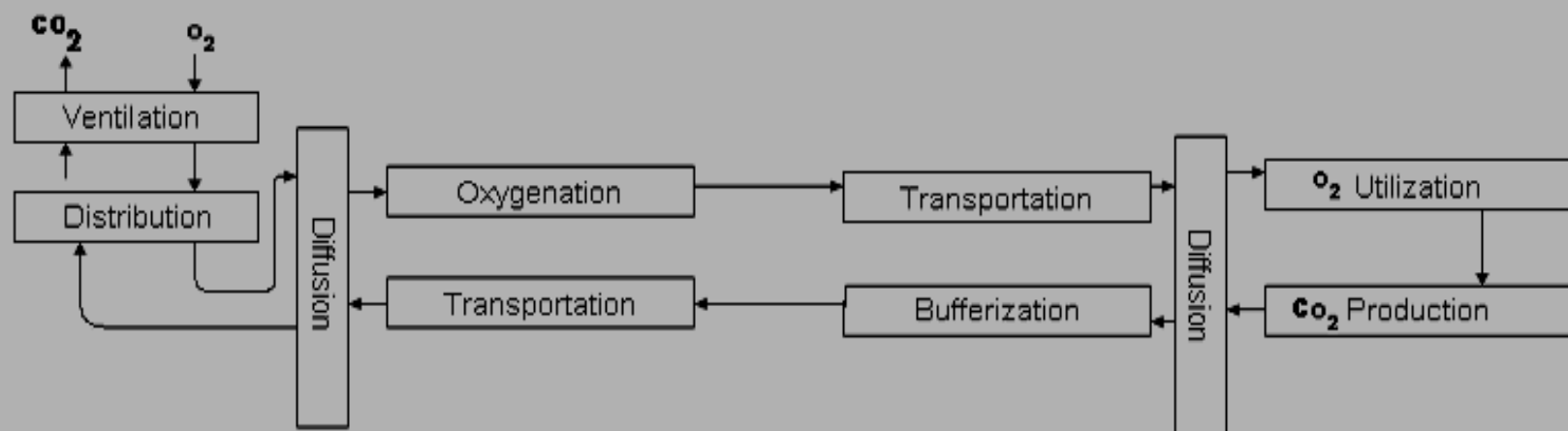
Stages of Respiration

1. **Pulmonary ventilation**: gas exchange between the atmosphere and lungs
2. **External respiration** -gas exchange between the lungs and blood (O_2 loading and CO_2 unloading).
3. **Transport of respiratory gases** –via movement of blood O_2 from the lungs is transported to the cell and tissues.
4. **Internal respiration** –gas exchange between the capillaries and the tissues (O_2 unloading and CO_2 loading).

Anatomical Design



Functional Processes



Operational Stages



Anatomical Consideration

1. Conducting Zone

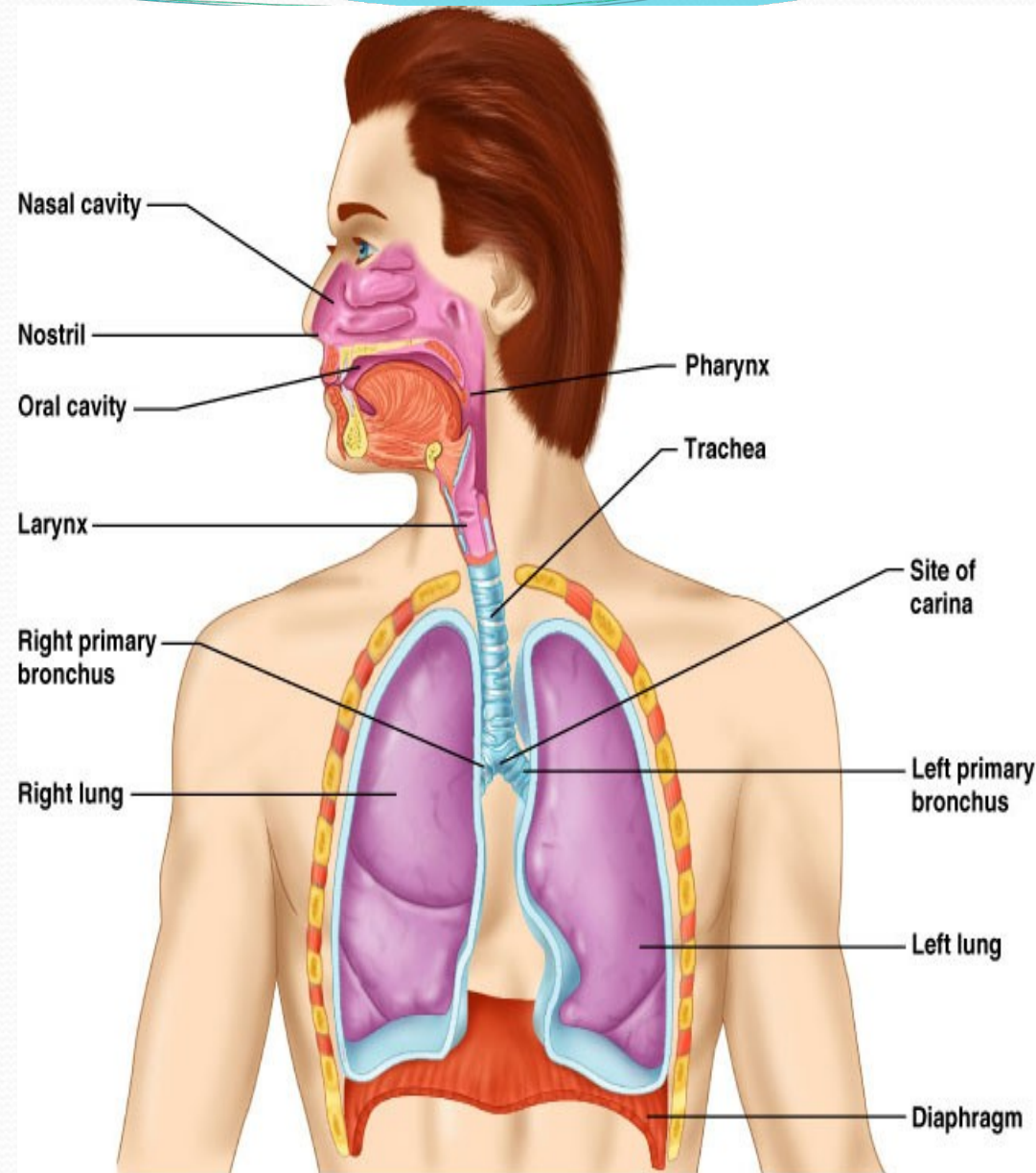
Rigid conduits for air to reach site of gas exchange

- Nose
- Nasal cavity
- Pharynx
- Larynx
- Trachea
- Bronchi

2. Respiratory Zone

site of gas exchange

- Respiratory bronchioles
- Alveolar ducts



Conducting Zone

Nasal Cavity

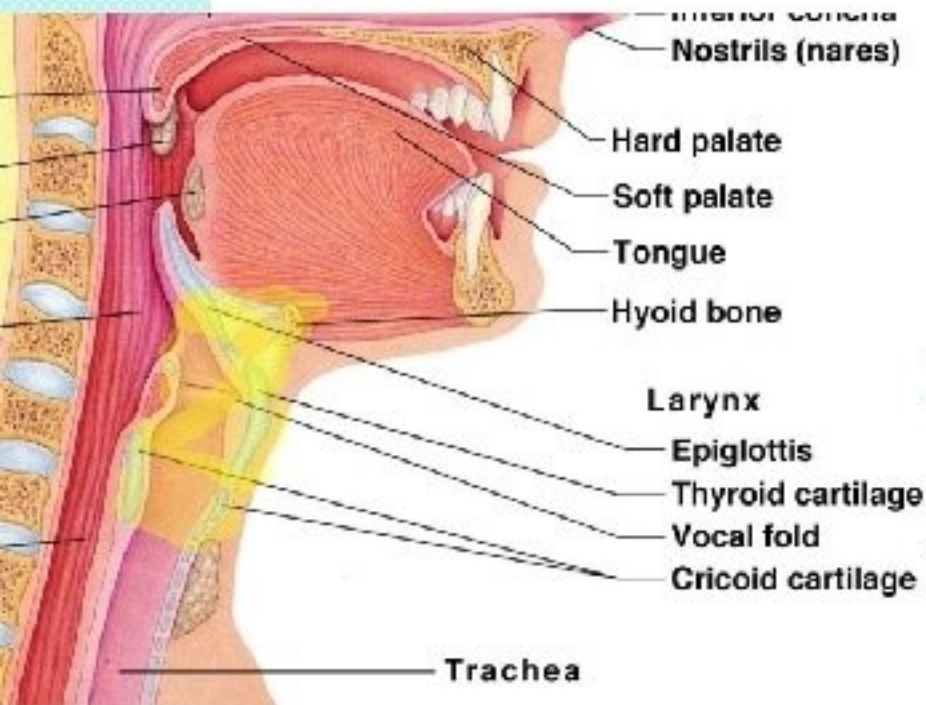
- Moistens and warms the air
- Filters inspired air
- Resonating chamber for speech

Pharynx

- Funnel-shaped tube of skeletal muscle
- Common passageway for: Food, Fluid and Air
- Located between:
 - ✓ Nasal cavity and mouth superiorly
 - ✓ Larynx and esophagus inferiorly
- Extends from the base of the skull to the level of the sixth cervical vertebra

Larynx (Voice Box)

- ☐ Routes air and food into proper channels
- ☐ Plays a role in speech
- ☐ Made of eight rigid hyaline cartilages and a spoon-shaped flap of elastic cartilage (epiglottis)
- ☐ Thyroid cartilage
 - Largest of the hyaline cartilages
 - Protrudes anteriorly (Adam's apple)
- ☐ Epiglottis
 - Protects the superior opening of the larynx
 - Routes food to the esophagus and air toward the trachea
 - When swallowing, the epiglottis rises and forms a lid over the opening of the larynx
- ☐ Vocal folds (true vocal cords)
 - Vibrate with expelled air to create sound (speech)
- ☐ Glottis—opening between vocal cords



Trachea

- Flexible and mobile tube extending from the **larynx into the mediastinum**
- Located anterior to the esophagus
- A fairly short tube (10-12 cm) with a diameter of about 2cm.
- Lined by respiratory epithelium.
- Musculo-cartilaginous structure (10cmx2cm) that branches into two bronchi.
- Composed of three layers
 - **Mucosa** – made up of goblet cells and ciliated epithelium
 - **Sub-mucosa** – connective tissue deep to the mucosa
 - **Adventitia** – outermost layer made of C-shaped rings of hyaline cartilage

Bronchi

- Conductive structures of a size down to ~1 mm are termed bronchi.
- Result of tracheal bifurcation
- Bronchi subdivide into secondary bronchi, each supplying a lobe of the lungs
- Bronchi are extension of trachea making the 1st to 11th generations while the terminal bronchioles make up the 12th -16th generations.
- Air reaching the bronchi is:
 - ✓ Warmed and cleansed of impurities
 - ✓ Saturated with water vapor
- At the transition from bronchi to bronchioles the epithelium changes to a ciliated columnar epithelium

CILIA

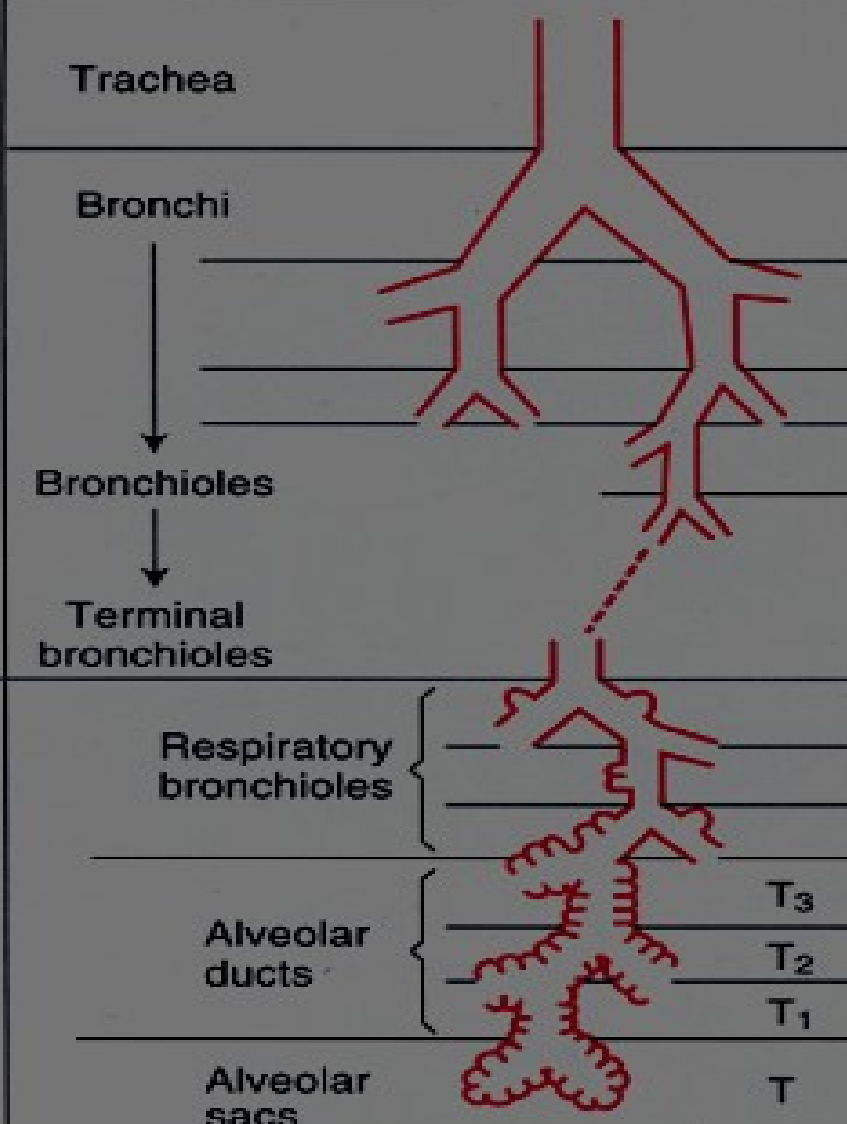
- **Hair-like projections** lining the respiratory passages to the end of the bronchioles
- Particulate matter sticking to mucus is moved by cilia to pharynx and swallowed thus keeping the lungs clean.
- The mucus in cilia forms an “**escalator**” called **mucus escalator**.
- This **defends** the air passages against bacterial infection.
- Ciliary activity reduced by noxious agents leads to lung infection.
- **A single cigarette** → non-motility of cilia for several hours
- Non-motility of cilia + secretion of mucus → partial or complete airway obstruction.

Respiratory Zone

- Defined by the presence of **alveoli**; begins as terminal bronchioles feed into respiratory bronchioles
- Respiratory bronchioles with alveoli in their walls make up the 17th-20th generations.
- Alveolar ducts making up the 21st-23rd generations terminate in alveolar sacs.
- At the transition from bronchi to bronchioles the epithelium changes to a ciliated columnar epithelium
- Respiratory bronchioles lead to alveolar ducts, then to terminal clusters of alveolar sacs composed of alveoli
- Approximately there are **300 million alveoli**:
 - ✓ Account for most of the lungs' volume
 - ✓ Provide tremendous surface area **for gas exchange**

ALVEOLI

- Have walls made up of:
 - 1) *Simple thin squamous epithelium- pneumocytes-type I*, major lining cells, where gas exchange takes place;
 - 2) *Type II pneumocytes:-* that are less in number and constitute thicker granulocytes responsible for the production of *surfactants*;
 - 3) *Type III pneumocytes-* large phagocytic macrophage cells found in alveolar cavities.
- ❑ These cells keep alveolar *surfaces sterile* by removing debris and microbes
- Are supplied with blood vessels, and lymphatics.
- Are connected by open pores.
 - Allow air pressure throughout the lung to be equalized.
- Form part of the *respiratory (alveolo-capillary) membrane*

Conducting zone	Trachea		Z
			0
	Bronchi		1
			2
			3
	Bronchioles		4
			5
	↓	16	
Transitional and respiratory zones	Respiratory bronchioles		17
			18
			19
	Alveolar ducts	T ₃	20
		T ₂	21
		T ₁	22
	Alveolar sacs	T	23

Idealization of the human airways according to Weibel. Note that the first 16 generations (Z) make up the conducting airways, and the last 7, the respiratory zone (or the transitional and respiratory zone).

Lungs

- There are two lungs
- Left lung – separated into upper and lower lobes
- Right lung – separated into three lobes
i.e Superior, Middle and Inferior
- They contain about 300 million alveoli whose surface area is about 70m² (about 40x the SA of the body).
- Each lung is surrounded by two layers of serous membrane (pleurae)-**visceral and parietal layers** separated by a small cavity filled with **pleural fluid**.
- This fluid assists in producing breathing movements by acting as a **lubricant**.

Blood and Nerve Supply to Lungs

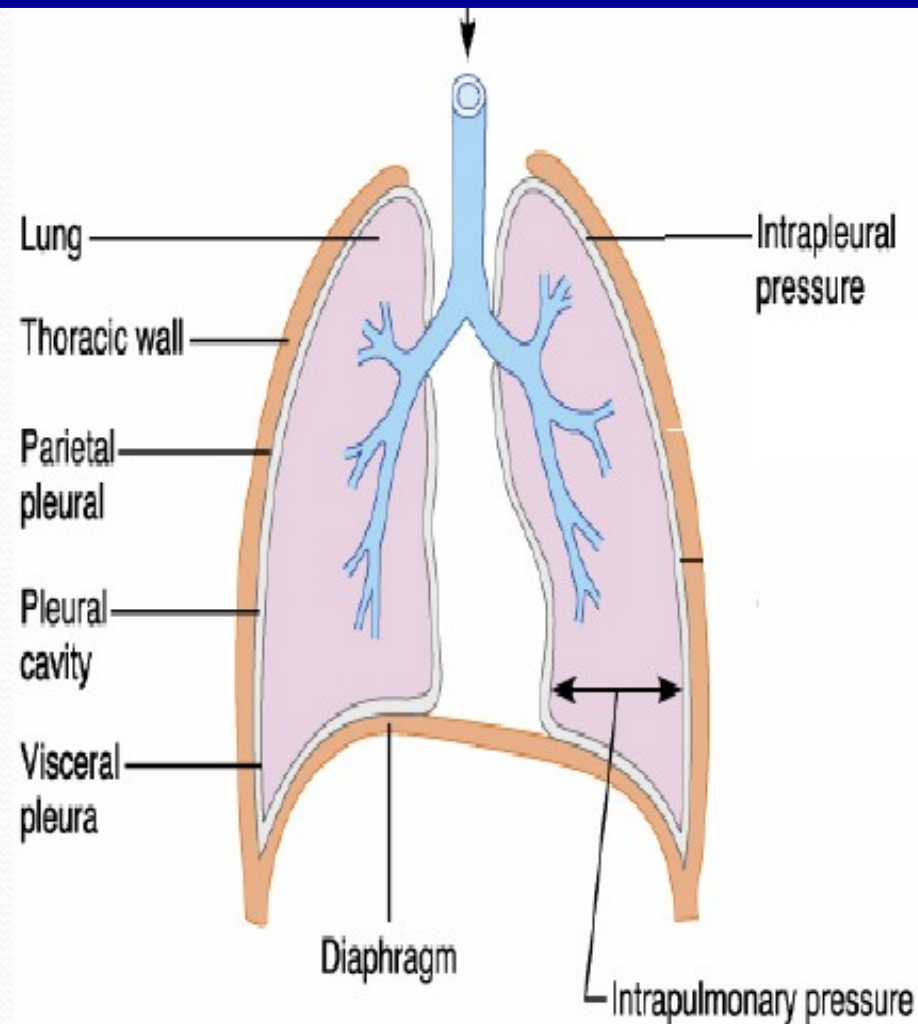
- Lungs are perfused by two circulations:
pulmonary and bronchial
- Pulmonary arteries – Conduct deoxygenated blood from right heart to lungs to be oxygenated
 - ✓ Branch profusely, along with bronchi
 - ✓ Ultimately feed into the pulmonary capillary network surrounding the alveoli
- Pulmonary veins – carry oxygenated blood from lungs to the left heart

Innervation

- Parasympathetic motor fibers (some sympathetic fibers), enter the lung through the pulmonary plexus on the lung root.
 - parasympathetic fibers –constrict the air tubes
 - sympathetic fibers –dilate air tubes
- Diaphragm: Phrenic nerves arising from C₃₋₅
- Intercostals: Intercostal nerves from thoracic segments

Lungs and Pleural Coverings

- **Pleural Coverings:**
 - Double layered serosa
 - Parietal pleura lines the thoracic wall
 - Pulmonary or visceral pleura which covers the lung surface
 - Pleural cavity is the space between the two layers
 - Pleural fluid fills the cavity



Breathing

- Breathing, or pulmonary ventilation, consists of two phases
 - **Inspiration** – air flows into the lungs (inhalation)
 - **Expiration** – gases exit the lungs (exhalation)

Pressure Relationships in the Thoracic Cavity

- Respiratory pressure is always described relative to atmospheric pressure (P_{atm})
- P_{atm} is pressure exerted by the air surrounding the body
 - ✓ Negative respiratory pressure is less than P_{atm}
 - ✓ Positive respiratory pressure is greater than P_{atm}

Pulmonary Ventilation

- A mechanical process that depends on volume changes in the thoracic cavity
- Volume changes lead to pressure changes, which leads to the **flow of gases to equalize pressure**.
- **Ventilation (V)** refers to the flow of air into and out of the alveoli, while **perfusion (Q)** refers to the flow of blood to alveolar capillaries.
- ❖ **What is the normal ventilation perfusion ratio?**
In a healthy individual, the V/Q ratio is **1 at the middle of** the lung, with a minimal spread of V/Q ratios from 0.3 to 2.1 from base to apex

Example: In **pneumonia** the alveoli are filled with exudate, impairing the delivery of air to the alveoli and lengthening the diffusion pathway for the respiratory gases

Inspiration

- ❖ The external intercostal muscles plus the diaphragm contract to bring about inspiration:
- Contraction of external intercostal muscles → elevation of ribs & sternum → ↑ antero-posterior dimension of thoracic cavity → lowers air pressure in lungs → moving of air into lungs.

Inspiration

- Contraction of **diaphragm** → movement of the diaphragm downward → ↑ vertical dimension of thoracic cavity → lowering of air pressure in the lungs → movement of air into the lungs:
- These leads the lungs to stretched and intrapulmonary volume increases
- Intrapulmonary pressure drops below atmospheric pressure (to -1 mm Hg)
- Intrapleural pressure drops below atmospheric pressure (to -6 mm Hg)
- Air flows **into the lungs until pressures equalizes**

Expiration

- Inspiratory muscles relax and the rib cage descends due to gravity
- Thoracic cavity volume decreases
- Elastic lungs recoil **passively** and intrapulmonary volume decreases
- Intrapulmonary pressure rises above atmospheric pressure (+1 mm Hg)
- Gases flow **out of the lungs down** the pressure gradient until pressures equalize

air enters

ribcage moves
up and out

lungs expand

diaphragm moves down

Inhalation

air leaves

ribcage moves
down and in

lungs get
smaller

diaphragm moves up

Exhalation

The Work (Mechanics) of Breathing

General: Forces → Movement of lungs and chest wall

Groups of muscles	Resistances to be overcome during inspiration (-)
1. Diaphragm (70% of TV)	Elastic resistance (65%) -elastic recoil of lungs -elastic recoil of thorax
2. Intercostals	
3. Auxilliary muscles: • Sternocleidomastoid • Scalene • Pectoralis • Latismus dorsi	Non-elastic resistance - airway resistance(20%) - tissue resistance of lungs & thorax (15%)

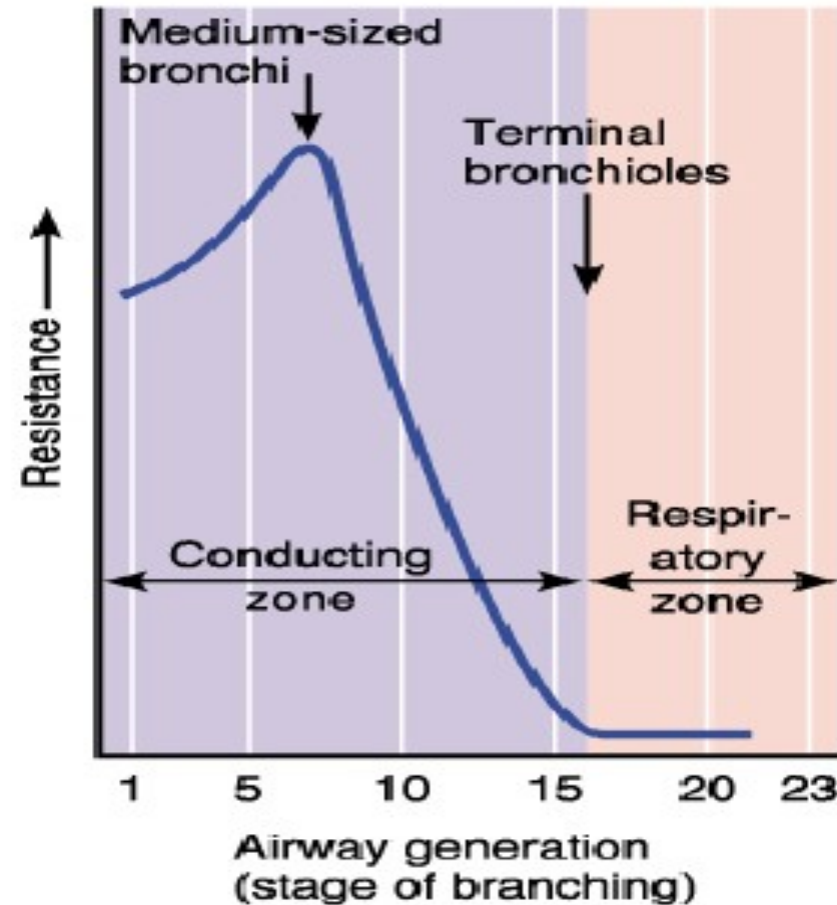
Physical Factors Influencing Ventilation:

Airway Resistance

- Friction is the major source of resistance to airflow
- The relationship between flow (F), pressure (P), and resistance (R) is:

$$F = \frac{\Delta P}{R}$$

➤ Maximum resistance is in **medium sized bronchi** then drops as cross sectional area increases



Physical Factors Influencing Ventilation:

Airway Resistance

- The amount of gas flowing into and out of the alveoli is directly **proportional to ΔP** , the pressure gradient between the atmosphere and the alveoli
- Gas flow is inversely proportional to **resistance** with the greatest resistance being in the medium-sized bronchi.
- As airway resistance rises, breathing movements become more strenuous.
- Severely constricted or obstructed bronchioles:
 - **Can prevent life-sustaining ventilation**
 - Can occur during **acute asthma attacks** which stops ventilation
- **Epinephrine** released via the sympathetic nervous system dilates bronchioles and reduces air resistance

Physical Factors Influencing Ventilation: Lung Compliance

- The ease (tendency) with which lungs can be expanded (Distensibility).
- **Compliance** is the force responsible for inspiration.
- Specifically, the measure of the change in lung volume that occurs with a given change in transpulmonary pressure
- Determined by two main factors
 - Distensibility of the lung tissue and surrounding thoracic cage
 - Surface tension of the alveoli

Factors That Diminish Lung Compliance

- Scar tissue or fibrosis that reduces the natural flexibility of the lungs
- Blockage of the smaller respiratory passages with mucus or fluid
- Reduced production of surfactant (surfactant decreases alveolar surface tension and prevents tendency of lung collapse)
- Decreased flexibility of the thoracic cage or its decreased ability to expand (restrictive lung diseases e.g., emphysema)

Pulmonary Ventilation Rate

- Respiratory rate and tidal volume determine the Pulmonary Ventilation rate
 - Pulmonary Ventilation is the term given to the **movement of air in and out of the lungs**; the pulmonary ventilation rate is defined as the tidal volume multiplied by the number of breaths taken per minute.
 - Pulmonary ventilation rate depends up on 2 factors:
 - **The size of each breath** (tidal volume: TV)
 - **The number of breaths/minute** (respiratory frequency: BR)
- $PV = BR \times TV$

Pulmonary Ventilation Rate

Example: suppose your tidal volume is 500 mL (0.5 liters) and you breathe 15 times/minute:

➤ Pulmonary ventilation rate = 15 breaths/min X 0.5 L/breath = 7.5 L/min

The Alveolar Ventilation Rate

Out of each breath approximately 150 cc is in the anatomical dead space (the conducting zone) where gas transport does not occur. In order to calculate the volume of air actually entering the alveoli the dead space volume must be subtracted.

$$AVR = \text{rate} \times [\text{tidal volume} - 150 \text{ cc}]$$

For example:

$$AVR = 12 \times [500 - 150 \text{ cc}] =$$

$$AVR = 12 \times 350 \text{ cc} = 4200 \text{ cc}$$

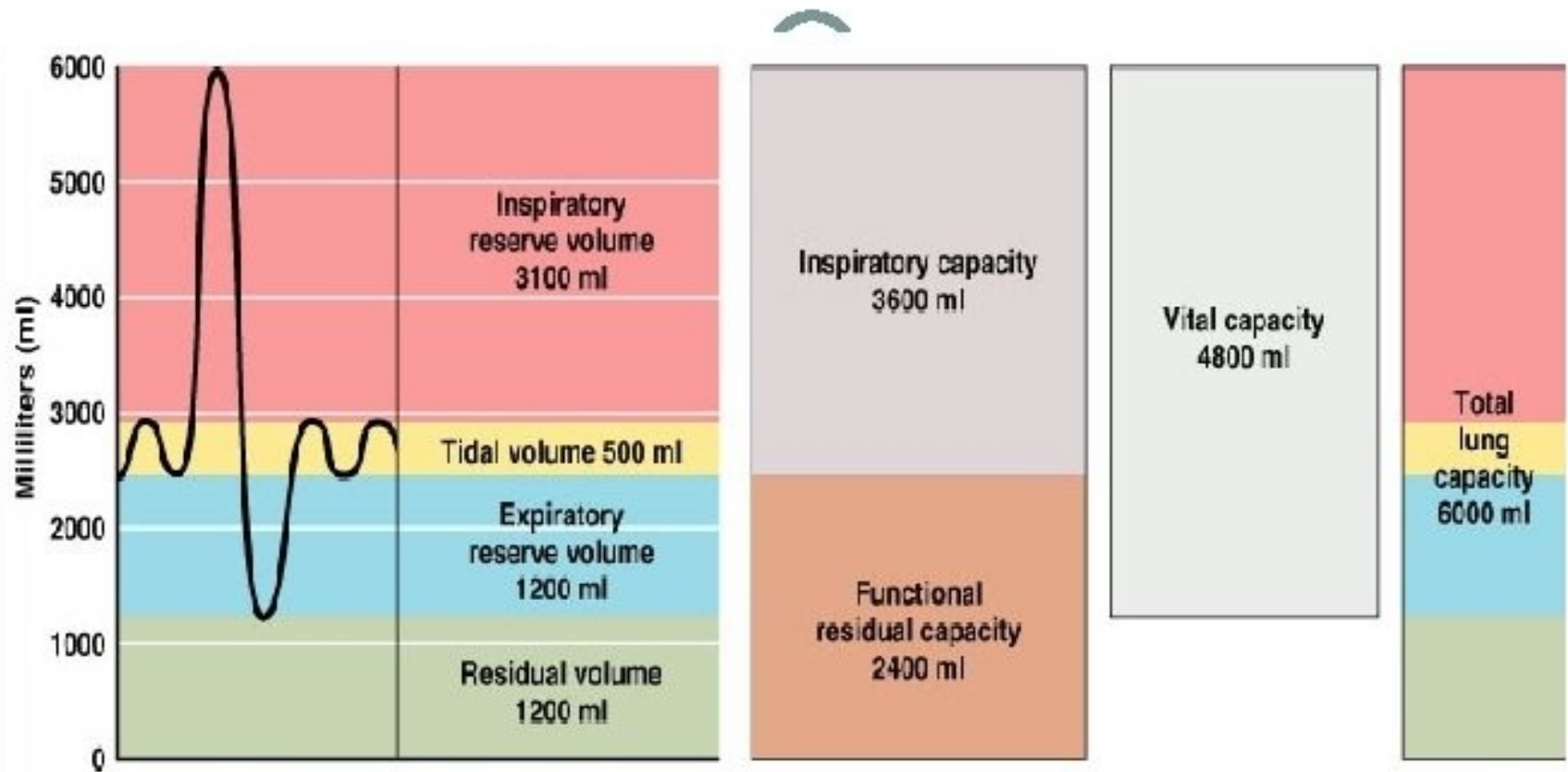
Respiratory Volumes

- There are four respiratory volumes and capacities.
- **Tidal volume (TV)** – air that moves into and out of the lungs with each breath (approximately 500 ml)
- **Inspiratory reserve volume (IRV)** – air that can be inspired forcefully beyond the tidal volume (3000 ml)
- **Expiratory reserve volume (ERV)** - air that can be expired forcefully beyond the tidal volume (1100ml)
- **Residual volume (RV)** – air left in the lungs after maximal expiration (1200 ml)

Respiratory Capacities

- **Vital capacity (VC)** – the total amount of exchangeable air (TV + IRV + ERV)
- **Inspiratory capacity** - the sum of TV and IRV
- **Functional residual capacity** - the sum of RV & ERV
- **Total lung capacity (TLC)** – sum of all lung volumes (approximately 5800 ml in males)

Respiratory Volumes & Capacities



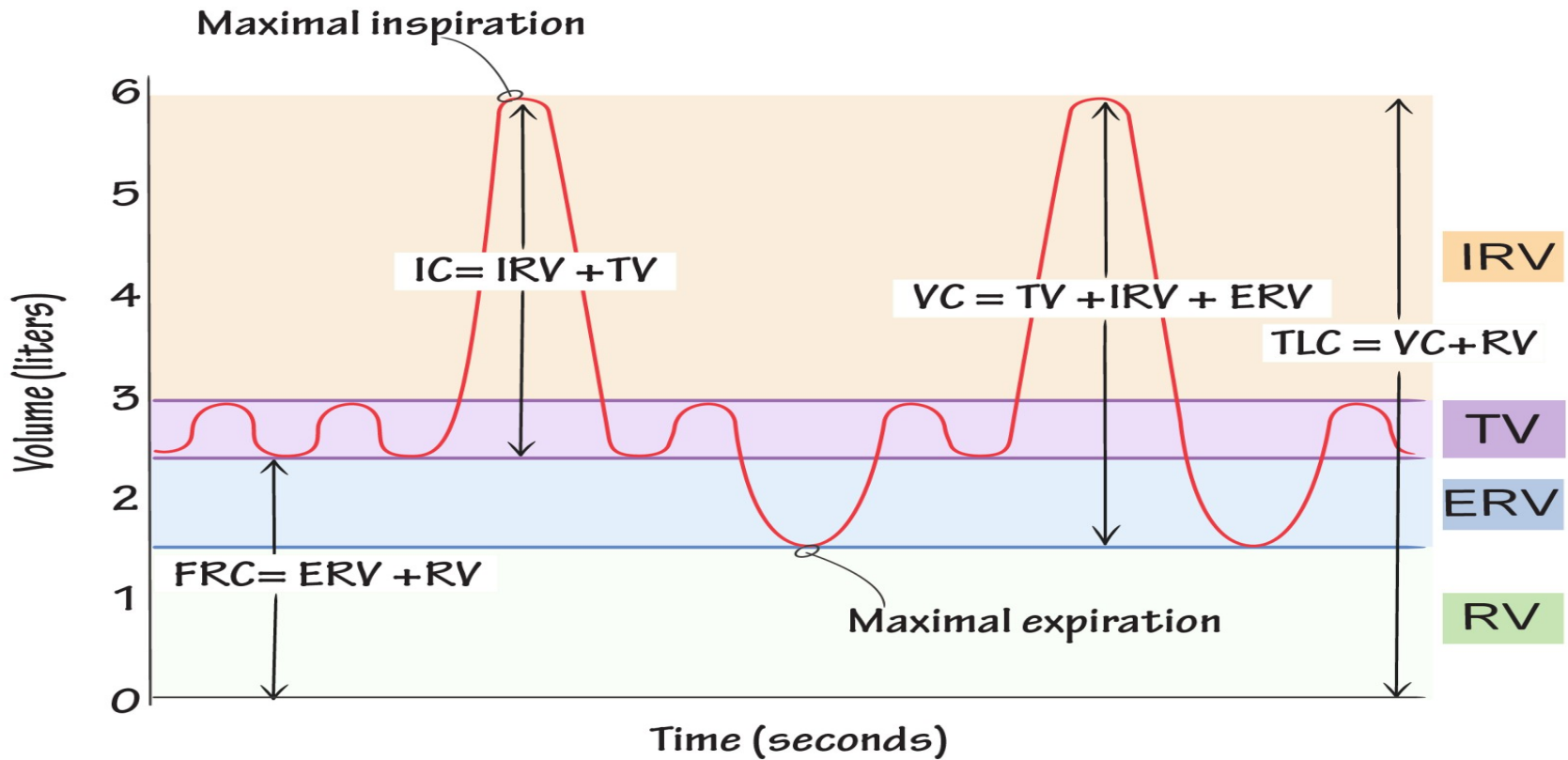
(a) Spirographic record for a male

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spirogram, is the most common measure of ventilatory lung function.

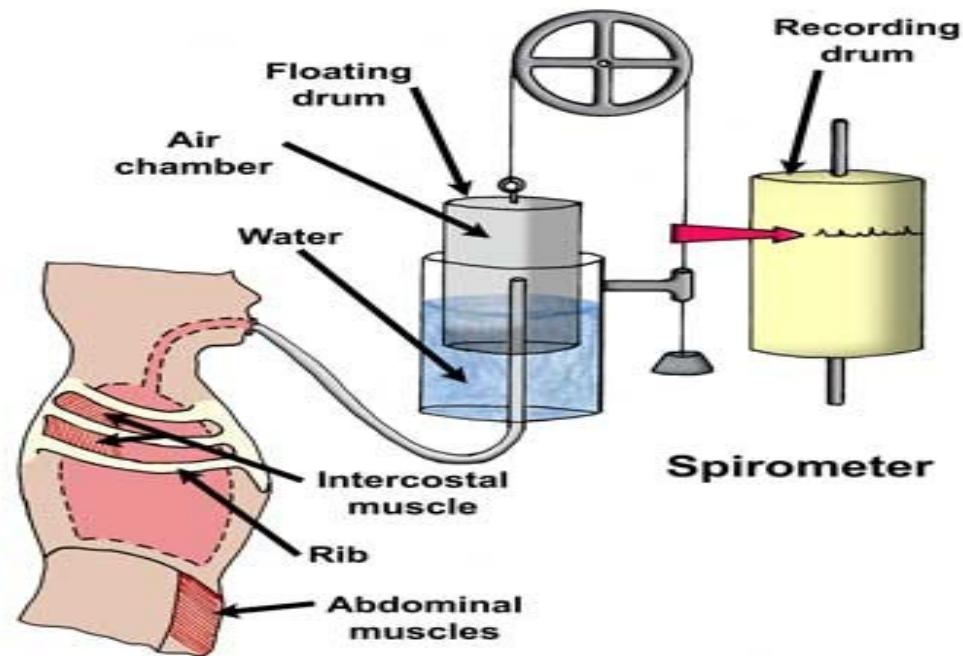
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Lung Volumes



Pulmonary Function Tests

- Spirometer – an instrument consisting of a hollow bell inverted over water, used to evaluate respiratory function



Non-respiratory Air Movements

- Mostly result from reflex action
 - Examples include: coughing, sneezing, crying, laughing, hiccupping, and yawning

External Respiration

Pulmonary Gas Exchange

- Factors influencing movement of oxygen and carbon dioxide across the **respiratory membrane** are:-
 - Partial pressure gradients and gas solubility
 - Matching of alveolar ventilation and pulmonary blood **perfusion**
 - Structural characteristics of the **respiratory membrane**

Partial Pressure Gradients and Gas Solubility

- The partial **pressure of oxygen** (PO_2) in capillaries is 40 mm Hg; the partial pressure in the alveoli is 104 mm Hg
 - This steep gradient allows oxygen partial pressures to rapidly diffuse from alveoli to capillaries

Partial Pressure Gradients and Gas Solubility

- Although carbon dioxide has a lower partial pressure gradient (alveolar capillaries = 45 mm, alveoli = 40 mm), but :-
 - It is 20 times more soluble in plasma than oxygen
 - It diffuses easily from capillaries to alveoli

Surface Area and Thickness of the Respiratory Membrane

- Respiratory membranes:
 - Are only 0.5 to 1 μm thick, allowing for efficient gas exchange
 - Very large surface area for gas exchange
 - If lungs have abnormal **fluid accumulation**, gas exchange is inadequate and oxygen deprivation results
 - Decrease in surface area with emphysema
→ Less surface area, less gas exchange

Inspired air:
 PO_2 160 mm Hg
 PCO_2 0.3 mm Hg



External respiration

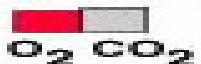
Blood entering alveolar capillaries:
 PO_2 40 mm Hg
 PCO_2 45 mm Hg



Pulmonary arteries

Systemic veins

Blood leaving tissue capillaries:
 PO_2 40 mm Hg
 PCO_2 45 mm Hg



Internal respiration

Tissues:
 PO_2 40 mm Hg
 PCO_2 46 mm Hg

Expired air:
 PO_2 120 mm Hg
 PCO_2 27 mm Hg



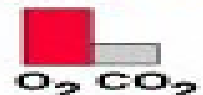
Blood leaving alveolar capillaries:
 PO_2 95 mmHg
 PCO_2 40 mmHg



Pulmonary veins

Systemic arteries

Blood entering tissue capillaries:
 PO_2 95 mmHg
 PCO_2 40 mmHg



Partial Pressure Gradients

Internal Respiration

- The **factors promoting gas exchange** between systemic capillaries and tissue cells are the same as those acting in the lungs.
 - The **partial pressures** and diffusion gradients are reversed
 - P_{O_2} in tissue is always lower than in systemic arterial blood (40mmHg & 95mmHg respectively)
 - P_{O_2} of venous blood draining tissues is 40 mm Hg and P_{CO_2} is 45 mm Hg & inside tissue is 46mmHg)

Regulation of Respiration

- Two major respiratory centers in **the brain stem** regulate respiration

1) Medullary respiratory center

- Consists of dorsal respiratory group (DRG) and ventral respiratory group (VRG)
- DRG contain mostly inspiratory neurons (I-neurons)
- Also responsible for rhythmic breathing
- VRG contain expiratory neurons (E-neurons)

Regulation of Respiration cont'

2) Pons respiratory center

- Influences output from medullary respiratory center
- **Pneumotaxic neurons** “switch off” I-neurons (limits duration of inspiration)
- **Apneustic neurons** prevent I –neurons from being switched off
- Pneumotaxic dominant over apneustic, allowing for smooth breathing

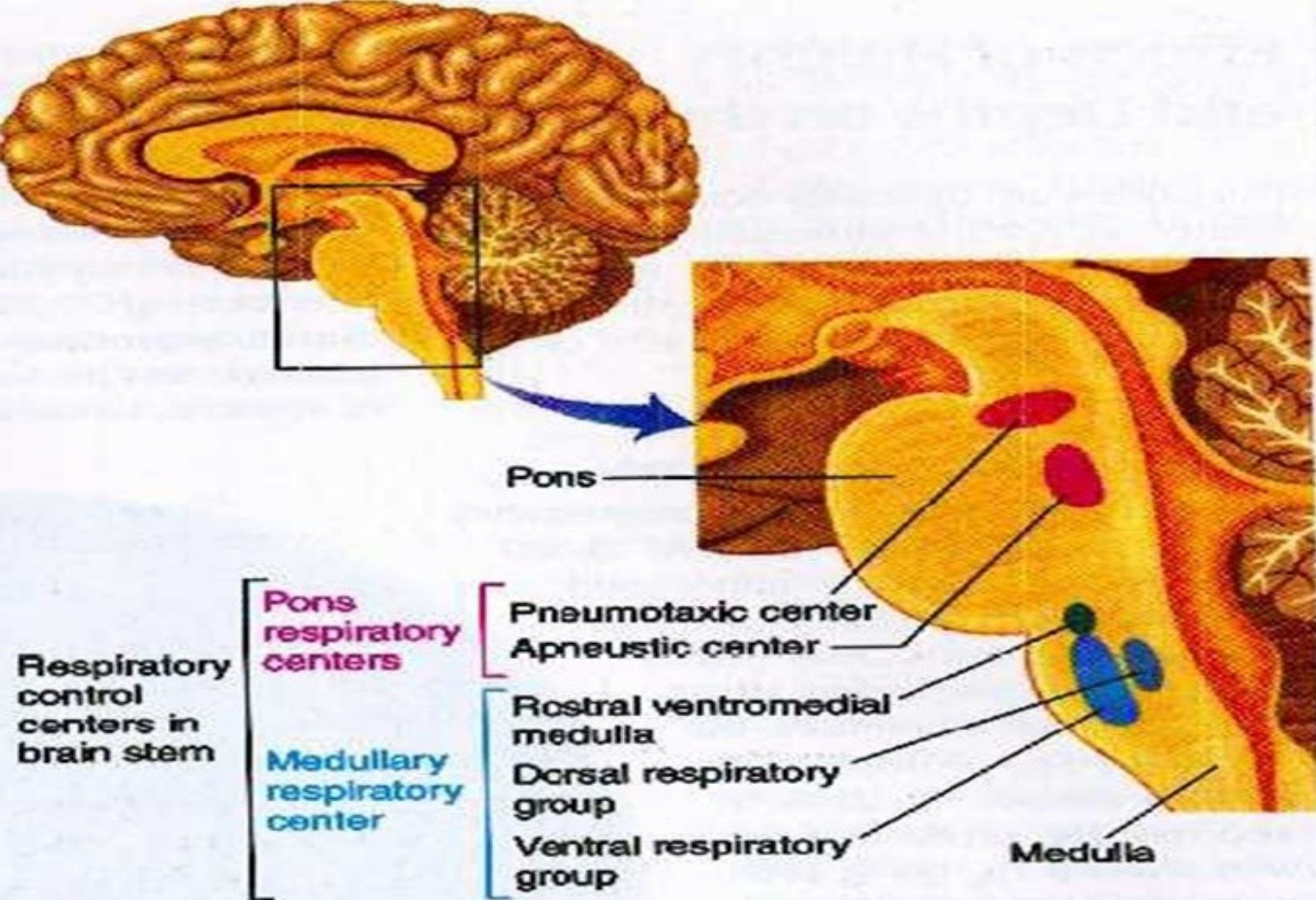


FIGURE
Respiratory control centers in the brain stem

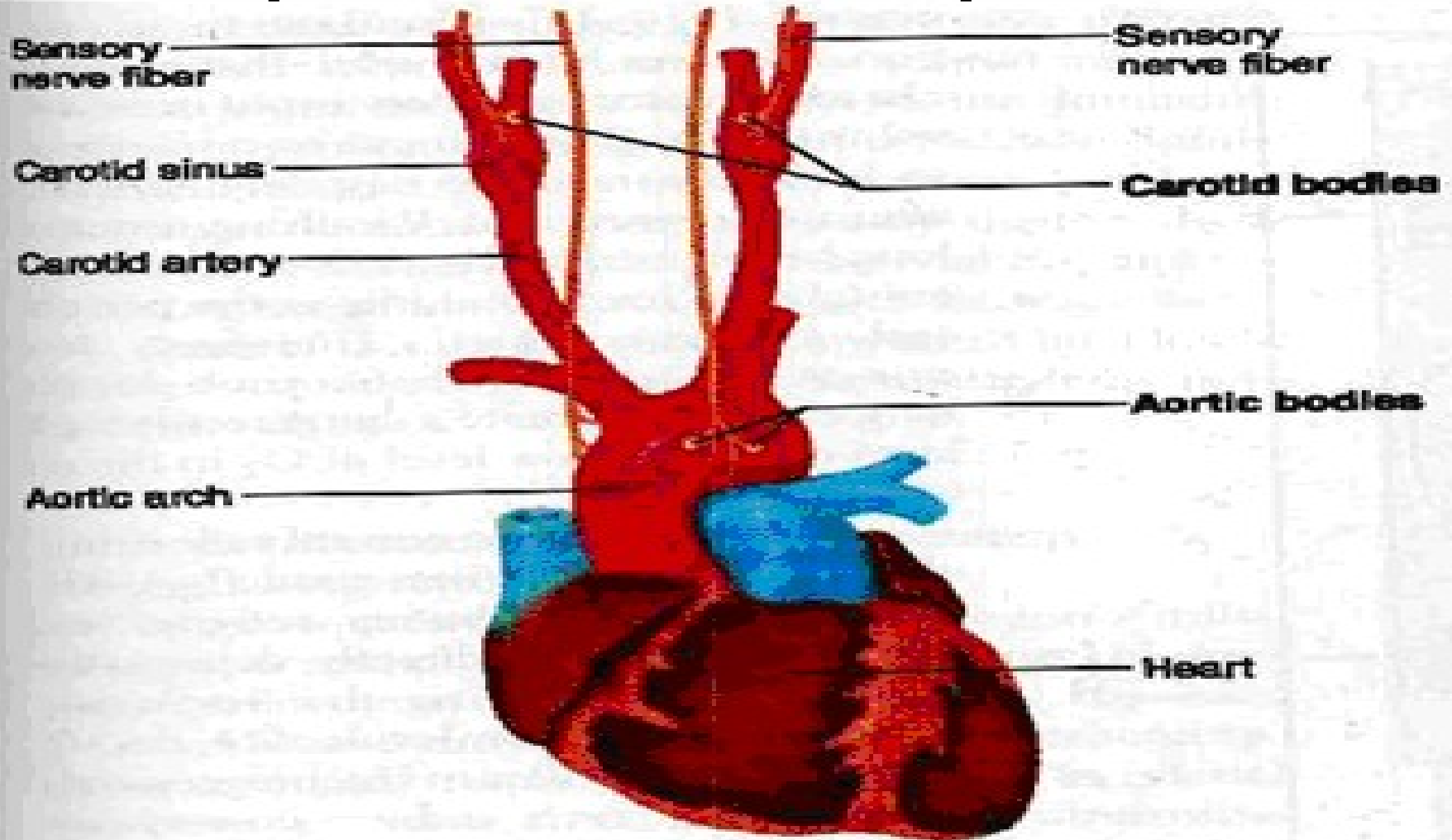
Regulation of Respiration cont'

- Control of ventilation by PO_2 , PCO_2 and H^+ is achieved via chemoreceptors (2 types)
 - 1) **Peripheral**- located in the carotid bodies and aortic bodies
 - 2) **Central**- located on the ventral surface of the medulla
- Controls breathing via nerve fibers to the respiratory control centers

Peripheral Chemoreceptors

- Sense changes in **arterial O_2 , CO_2 and H^+**
- **$\uparrow PCO_2$** $\rightarrow$ chemoreceptor $\rightarrow$ generation of AP $\rightarrow$ conduction of AP via sensory neurons $\rightarrow$ respiratory control centre $\rightarrow$ respiratory muscle $\rightarrow$ $\uparrow$ ventilation (CO_2 blown off) $\rightarrow$ $\downarrow PCO_2$
- **$\uparrow H^+$ (lactic acids)** $\rightarrow$ chemoreceptor $\rightarrow$ respiratory control centre $\rightarrow$ $\uparrow$ ventilation $\rightarrow$ $\downarrow PCO_2 \rightarrow \downarrow H^+$
 - Control of respiration in mammals is regulated by changes in PCO_2 (not PO_2).
 - Peripheral O_2 chemoreceptors do not contribute in regulating normal ventilation unless arterial PO_2 falls below 60 mm Hg.
 - Peripheral O_2 , CO_2 and H^+ chemoreceptors are weakly responsive and play a minor role in controlling respiration.

Peripheral Chemoreceptors



■ FIGURE

Location of the peripheral chemoreceptors

The carotid bodies are located in the carotid sinus, and the aortic bodies are located in the aortic arch.

Central chemoreceptors

- **Most important** regulator of ventilation
- Central chemoreceptors are found in the **medulla**.
- Do not monitor changes in PCO_2 directly
- Respond to changes in **CO_2 -induced** production of **H^+** in cerebrospinal fluid (brain interstitial fluid)
- Blood-brain barrier allows the diffusion of CO_2 but is impermeable to H^+

